HL-CYB-001

Lab implementation guide: five phases

Enterprise Zero Trust Security Architecture

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PLANNED - operational controls NOT TESTED

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T6 baseline, 2026-11-06. Implementation PLANNED; operational controls NOT TESTED; owner user acceptance testing (UAT) PENDING.

This guide supports the four-week assignment. Read the architecture, security controls and test plan together. T6 changes documents only; none of the implementation examples below has been executed.

1. How to execute a future approved stage

Before each stage record owner approval/date, exact assets, permitted changes, validation, private evidence handling and rollback in a new draft task report. Verify current installed-version guidance, media integrity and lawful terms. Five phases organize work; they retain the separate A-L approvals in AGENTS.md.

Table 1. How to execute a future approved stage

Phase	Existing approval stages retained	Week
1: Infrastructure and Network	A readiness, B host, C networking, D virtual firewall; each separately	1
2: Identity and Endpoint Security	E directory, F endpoint/server baselines	2
3: Zero Trust Access Controls	G simple CA, H application roles, I Tailscale	2-3
4: Monitoring, Automation and Recovery	J monitoring; L automation/recovery portion	3-4
5: Validation and Final Submission	K validation; remaining L closeout portion	4

T6 replaces old stage quantities, not approval gates. Only the explicitly approved automation/recovery portion of L may precede K; prior approval does not authorize remaining changes. Allow about 8-12 focused hours weekly after readiness/approvals. Pause dependent work on delivery, isolation, activation or recovery gaps and revise dates.

All implementation examples are **proposed future commands/configuration**, with targets stated. Set symbolic variables privately to reviewed lab-only values outside Git. Stop on uncertain assets/disks, unexpected protected connectivity, secret output, missing rollback or failed safety checks. Use approved console isolation and retain private evidence. Never execute a guide merely because it exists.

2. Phase 1: Infrastructure and Network

2.1. Prerequisites

Stage A read-only approval first. Existing owner-described hardware: Dell Precision 5820 (W-2125, 80 GB RAM, 512 GB SSD), Lenovo laptop, HP desktop, test laptops and Google Wifi; thermal paste available.

Ordered: Xeon W-2145, Samsung 1 TB SSD, Sabrent PCIe adapter, Intel I350-T4 NIC, GL.iNet Beryl AX and TP-Link managed switch. Reference: latest owner procurement plan and 2026-11-06 procurement correction. **Delivered:** unconfirmed; expected November 7 for non-CPU items and November 12, 2026 for CPU. **Installed:** no T6 installation evidence or assumption that Proxmox/other lab software exists. **Validated:** no hardware-readiness/operational-lab evidence. Do not reorder these items.

Before installation confirm delivery, physically inspect parts, verify compatibility, inventory existing data/services, establish backup/recovery readiness and obtain exact owner approval for destructive changes. Verify CPU/cooler/firmware and SSD/adapter/slot compatibility. Preserve original Windows disk and existing services; shutdown/migration requires separate approval. Extra hardware does not mandate physical OPNsense, VLANs or extra zones.

2.2. Required tools

Existing inventory/firmware/storage screens, physical console, private parts/disk/cable ledger, verified official media/checksums, private credential store and recovery media. No new inspection software during read-only A.

2.3. Implementation instructions

1. **A:** inspect CPU/RAM/virtualization, disk identity/contents/free space, NICs and console. Compare supplied/observed facts and date each hardware status separately. Do not open an installer.
2. Inventory current services/data and availability needs. Verify backup/recovery; exclude protected workloads before any approved host transition.
3. Recalculate the capacity ledger: four VMs, 20 GiB guest RAM/176 GiB disks, 12 GiB host RAM and 80 GiB free disk reserved, plus media/restore space. Issue no-go if it cannot fit.
4. Map cables/bridges and administrator adapters. No forwarding via Wi-Fi, another NIC or VPN; no workload bridge physical uplink. Keep external access disconnected.
5. **B:** after exact-disk/destructive-install approval and verified recovery, boot official verified Proxmox media at console. Recheck disk identity immediately before selection; never use the preserved Windows disk or unapproved substitute.
6. Configure one host with named scoped administration, supported MFA/recovery and isolated management. Record build/updates/storage and console recovery. Root is for necessary setup/recovery. No cluster.
7. **C:** create vmbr0 Management, vmbr10 Corporate and vmbr20 Restricted. Only vmbr0 has the verified isolated physical admin segment/host IP. Save prior settings and apply from console.
8. **D:** create OPNsense (2 vCPU/4 GiB/16 GiB), one NIC per zone. Match recorded virtual MAC/interface identities; install verified media. Optional WAN remains disconnected; no inbound forwarding.
9. Apply FF-01 through FF-08, replacing broad installer permits with exact source/destination/service allows. Keep the named management allow path before narrowing rules.

10. Inspect automatic/floating/group/anti-lockout rules, states, NAT, DNS, IPv6, firewall-originated traffic and UPnP/NAT-PMP. Save encrypted private config. Internet dependencies await approved independently isolated egress.

Proposed read-only inventory on **approved Windows candidates**, output private:

```
Get-CimInstance -ClassName Win32_ComputerSystem
Get-CimInstance -ClassName Win32_Processor
Get-NetIPConfiguration -All
```

Use version-matched Proxmox/OPNsense guidance; no shortcut disk-selection or packet-filter-disable command.

2.4. Expected results

Verified readiness and, only after B-D approvals, one recoverable host/firewall, three zones and no protected-network path. Hardware states reflect observations.

2.5. Verification steps

FWT-01/FWT-02: compare cables/bridges/NICs/routes/IPv6/aliases and effective policy with permitted/denied bounded flows to known listening **lab-owned mocks**; inspect logs/counters. Repurpose approved guests/console, no permanent extra VMs. Never probe home/production. Verify named administration, ordinary-user denial and console recovery.

2.6. Troubleshooting

Lost administration: inspect interface mapping/specific allow from console. Unexpected permit: disconnect approved egress, inspect states/automatic rules/forwarding. Uncertain disk/compatibility/backup blocks installation; no substitute disk/home uplink.

2.7. Rollback procedures

A is read-only. For B-D restore specific saved host/firewall/network settings from console; keep egress disconnected until reverified. Destructive reinstall/hardware changes need explicit rollback coverage. Preserve data/evidence.

3. Phase 2: Identity and Endpoint Security

3.1. Prerequisites

Phase 1 verified; E/F approvals; supported lawful Windows Server/Windows/Linux media, approved update/activation path, private DNS/address/time and recovery custody.

3.2. Required tools

Proxmox console; Server Manager, Active Directory Users and Computers, DNS/Group Policy Management; supported Linux/Windows firewall/settings tools and baseline checklist.

3.3. Implementation instructions

1. Create DC1 in Corporate (2 vCPU/6 GiB/64 GiB), WIN1 in Corporate (2/6/64), APP1 in Restricted (2/4/32). Each has one zone NIC; check supported CPU/disk/virtual TPM needs before installation.
2. Install approved OSs, record builds/update status/evaluation deadlines and configure lab DNS/time. Missing safe activation remains a dependency.
3. On DC1 set fixed private address/name, install AD Domain Services/DNS and create ad.lab.test. Use secure Directory Services Restore Mode prompts/private custody; no corporate trust/cloud join/sync.
4. Create user/privileged/computer organizational units. Add standard **ava**, **eli**, **uma**, plus one separate named privileged admin. Three project security groups: **LabReaders** (ava, eli), **LabMaintainers** (eli), **LabAdmins** (admin). Delegate only required rights. Secure built-in/service accounts separately; they are not extra project personas.
5. Join WIN1 to the lab domain; use lab DNS only. Add APP1's DNS record; APP1 uses approved lab DNS/time without a domain join.
6. Pilot Group Policy for host firewall, supported antivirus, audit logon/account changes (including DC Kerberos Authentication Service success/failure), sensible password/lockout and standard/admin separation. Protect recovery accounts; no lockout/password-attack experiment.
7. Enable APP1 host firewall; HTTPS only from approved sources, administration only from the dedicated path. Disable unnecessary listeners/forwarding; retain logs.
8. Inspect resultant policy/local Administrators. Record supported directory recovery as a dependency; snapshots do not prove AD disaster recovery.

Proposed **DC1 elevated PowerShell**, E approval only; changes configuration and may reboot:

```
Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools
Install-ADDSForest -DomainName 'ad.lab.test' -InstallDNS
```

Check selected-version parameters/prompts in Microsoft guidance. Proposed read-only **WIN1 member** check: `Test-ComputerSecureChannel -Verbose`; not a DC health test.

3.4. Expected results

One directory, three standard users/groups, separate privileged admin, Windows member and hardened Linux server. AD Windows sign-in and app permissions are verified separately.

3.5. Verification steps

FWT-03/FWT-04: compare directory/DNS/groups with WIN1 standard logon, name resolution, resultant policy and harmless privileged-action denial. Inspect firewall/patch/antivirus state and trust/cloud/resolver settings; never contact protected assets.

3.6. Troubleshooting

Join/logon/policy failure: inspect lab DNS/service records, time and AD/host rules. WIN1/DC1 share Corporate, so OPNsense does not mediate their traffic. No corporate DNS/credentials or broad permits.

3.7. Rollback procedures

Restore only changed pilot policy/group/host settings, preserving console/admin access. Forest rebuild/demotion or directory restore needs a separately approved supported plan; do not casually revert DC snapshots.

4. Phase 3: Zero Trust Access Controls

4.1. Prerequisites

Phase 2 verified; G/H/I approvals; app.ad.lab.test, frozen six-case matrix and private keys/passwords/trust scope. Tailscale requires approved independently isolated Internet, separate lab tailnet/identities, supported MFA/recovery and current terms; otherwise remote checks stay NOT TESTED.

4.2. Required tools

Supported Apache HTTP Server 2.4 file/group authorization and TLS, OpenSSL 3.x, lab browser/clients, AD group view and Tailscale console/client. Verify distribution package/config paths and installed help before approved installation.

4.3. Implementation instructions

1. On APP1 create root-owned synthetic pages in /srv/labapp/reports and /srv/labapp/maintenance. Maintenance-page reading is the restricted engineering workflow; expose no shell/real operational change.
2. **G:** create one encrypted CA key/public root on an approved isolated signing workstation; keep files offline outside Git. Transfer reviewed CSR/public certificates via approved media. No CA VM/separate issuer.
3. Generate APP1 key/CSR, restrict key permissions, inspect/sign the CSR offline with CA:FALSE, serverAuth, proper key usage and DNS SAN app.ad.lab.test. Return public leaf only.
4. Import public CA into approved lab clients; inspect hostname/dates. No home/production trust change or DC time manipulation.
5. **H:** create Apache accounts ava/eli/uma with prompted app passwords different from AD. Restrict hashed password/group files outside the document tree. Manually mirror LabReaders/LabMaintainers; LabAdmins is infrastructure-only. This is not AD SSO/automatic synchronization.
6. Configure one HTTPS virtual host, correct key/leaf, no plain HTTP/public access, listing off, two protected paths and secret-free access/error logs. Syntax-check before supported activation/reload.
7. Execute the six FWT-03 cases. Remove eli from app LabMaintainers and corresponding AD role; authenticated and fresh maintenance requests must fail. Record time/cache treatment. Restore only under approval.
8. **I:** confirm separate lab tailnet/MFA/recovery, replace permissive defaults, inspect all grants/ACLs/tag owners, then enroll APP1/one approved client. Grant exact APP1 TCP 443 only.
9. Keep subnet advertisements, exit nodes, public sharing, Tailscale SSH/Funnel/Serve and forwarding off. Inspect effective overlay host rules/vendor precedence; OPNsense alone cannot prove overlay restriction.
10. Map app.ad.lab.test privately to APP1 overlay address on the remote client; local clients use zone address. Test permitted/unauthorized/wrong listening port, remove client authorization/grants, and verify existing/fresh session denial. Record actual time and stop on broader access.

Proposed **offline signer**, G approval; private path variables, interactive encrypted-key prompt:

```
openssl req -x509 -newkey rsa:3072 -sha256 -days 365 -keyout "$LabCaKeyPath" -out
"$LabCaCertPath" -subj "/CN=HL-CYB-001 Lab CA" -addext "basicConstraints=critical,CA:TRUE"
-addext "keyUsage=critical,keyCertSign,cRLSign"
```

Proposed **APP1** key/CSR; restrict filesystem access to the unencrypted service key:

```
openssl req -new -newkey rsa:2048 -noenc -keyout "$LabAppKeyPath" -out "$LabAppCsrPath" -subj
"/CN=app.ad.lab.test"
```

Private reviewed leaf extension file on **offline signer**:

```
basicConstraints=critical,CA:FALSE
keyUsage=critical,digitalSignature,keyEncipherment
extendedKeyUsage=serverAuth
subjectAltName=DNS:app.ad.lab.test
```

Proposed **offline signer**, signing and inspection:

```
openssl x509 -req -in "$LabAppCsrPath" -CA "$LabCaCertPath" -CAkey "$LabCaKeyPath"
-CAcreateserial -CAserial "$LabSerialPath" -days 90 -sha256 -extfile "$LabLeafExtensionsPath"
-out "$LabAppCertPath"
openssl verify -CAfile "$LabCaCertPath" -verify_hostname app.ad.lab.test "$LabAppCertPath"
openssl x509 -in "$LabAppCertPath" -noout -dates -ext subjectAltName
```

Browser TLS is a second observation. Certificate revocation infrastructure is excluded; app/device revocation remains mandatory.

Proposed **APP1** password creation: `htpasswd -B -c /etc/labapp/users ava` once, then `htpasswd -B /etc/labapp/users eli` and similarly uma. -c overwrites records; never reuse it to add users. Passwords stay interactive. Private group file:

```
LabReaders: ava eli
LabMaintainers: eli
```

Proposed **APP1 Apache 2.4 fragment**, after modules/paths are verified; add Listen 443 once and disable default/plain-HTTP sites:

```
<VirtualHost *:443>
    ServerName app.ad.lab.test
    DocumentRoot "/srv/labapp"
    SSLEngine on
    SSLCertificateFile "/etc/labapp/app.crt"
    SSLCertificateKeyFile "/etc/labapp/app.key"
    SSLProtocol -all +TLSv1.2 +TLSv1.3
    <Directory "/srv/labapp">
        Options -Indexes
        AllowOverride None
        AuthType Basic
        AuthName "HL-CYB-001"
        AuthBasicProvider file
        AuthUserFile "/etc/labapp/users"
        AuthGroupFile "/etc/labapp/groups"
        AuthzSendForbiddenOnFailure On
        Require all denied
    </Directory>
    <Directory "/srv/labapp/reports">
        Require group LabReaders
    </Directory>
    <Directory "/srv/labapp/maintenance">
        Require group LabMaintainers
    </Directory>
</VirtualHost>
```

Default AuthMerging Off lets child authorization replace parent denial; inspect inherited/global policy. No broad Require valid-user. Configure package-appropriate logs. Proposed APP1 check: `apachectl configtest` (wrapper may differ); Syntax OK does not prove authorization. Proposed enrolled-device inspection: `tailscale status`, output private. See primary references.

4.4. Expected results

One CA/leaf, two protected paths, six role cases, app revocation and separate remote permit/deny/revoke evidence. No live AD integration/universal MFA/enterprise certificate lifecycle claim.

4.5. Verification steps

FWT-03/FWT-05/FWT-06: compare page behavior to effective roles/logs; allowed content renders, unauthorized content does not, with intended denial (normally 403 here). Inspect chain/name/dates and untrusted-client failure. Inspect vendor/host rules plus actual wrong-port denial to known listening lab service. Revoke existing/new requests, recording state/cache handling.

4.6. Troubleshooting

TLS: check SAN/chain/dates/trust/private DNS; clicking through warnings is not PASS. App: check spelling, group file, inherited rules/logs/browser credentials; AD alone does not update app roles. Remote: inspect lab tailnet, grants, overlay rules and approved egress; no broad routes/allow-all.

4.7. Rollback procedures

Restore saved app/group settings, syntax-check/reload. Remove trust only from approved clients as needed. Disable remote grant/device, stop client, restore lab DNS/routes and close temporary egress under approval. Preserve failures; exposed keys invoke the owner's authorized credential process.

5. Phase 4: Monitoring, Automation and Recovery

5.1. Prerequisites

J and specific L automation/recovery approvals, private records, one operation/target, encrypted off-compute storage, disconnected restore bridge and temporary capacity. Approval identifies exact APP1 targets; no AD/host disaster recovery implied.

5.2. Required tools

Firewall live log, DC1 Event Viewer, Apache logs, existing PowerShell/Python, Proxmox Backup/Restore UI, encrypted storage, hashes and independent clock. No SIEM/collector VM.

5.3. Implementation instructions

1. Use firewall, DC1 Security and Apache access/error sources. Two sources require approved selection covering all five categories. Restrict log access/export, rotate within disk capacity and record UTC/time differences.
2. Generate five benign events: directory success, one single failed logon in disposable synthetic context, app allow, authenticated app denial and firewall denial to a known lab listener. No password attack.
3. Record source/type/context, source/observer times and private evidence ID. Investigate one denial: requested action, rejecting rule/group, working control, correlated evidence and conclusion. On DC1 use the approved account-logon audit view: Kerberos 4768 success and 4771 failed pre-authentication are candidate records; confirm actual event generation from WIN1 and avoid cached/offline logon as the success observation. No enterprise latency/retention target.
4. Define one **local WIN1 capacity-summary operation** using PowerShell/Python. Review hostname/domain allowlist, read-only inputs, private output and invalid target/path rejection before future coding. No remote credentials; one JSON/CSV outside Git.
5. Run the future approved operation, compare output to Windows storage UI and repeat. Measurements may change; only intended output may be replaced, with no configuration/account/network change. Test rejection safely.
6. Define APP1 backup manifest: pages, Apache config, hashed accounts/groups, server key/leaf and public root. Use one consistent full-VM backup on **encrypted off-compute storage**; ordinary Proxmox archive format alone is not encryption.
7. Stop APP1 cleanly for the approved backup mode/target. Verify completion/readable manifest/private recovery-key access. Preserve the only valid backup; same-disk snapshots do not qualify.
8. Restore to an approved temporary VM ID with NICs disconnected, original APP1 off and copied Tailscale inactive. This copy serves the required test only; never connect duplicate identities.
9. Measure restore start to usable-service/data checks. By console/isolated client compare hashes/content, HTTPS and allowed/denied roles. If needed repurpose WIN1 onto a disconnected test bridge under approval, then restore attachment. No Management/home connection.
10. Record age, actual recovery time, scope/verdict/exclusions. APP1 restore does not prove AD/CA/host recovery or enterprise RPO/RTO. Leave copy disconnected; resuming original/deleting copy needs approved cleanup.

Proposed **WIN1 only, L automation approval**, save as a reviewed private script. Pass an existing approved private output directory outside every Git checkout. This one operation reads local capacity and replaces only its named summary; it changes no lab configuration.

```
param([Parameter(Mandatory)][string]$OutputDirectory)
$system = Get-CimInstance -ClassName Win32_ComputerSystem
if ($system.Name -ne 'WIN1' -or $system.Domain -ne 'ad.lab.test') {
    throw 'Target is outside the approved lab operation.'
}
$directory = Get-Item -LiteralPath $OutputDirectory -ErrorAction Stop
if (-not $directory.PSIsContainer) { throw 'Output must be a directory.' }
$cursor = $directory
while ($null -ne $cursor) {
    if (Test-Path -LiteralPath (Join-Path $cursor.FullName '.git')) {
        throw 'Keep the capacity record outside Git.'
    }
    $cursor = $cursor.Parent
}
$capacity = Get-CimInstance -ClassName Win32_LogicalDisk -Filter 'DriveType=3' |
    Select-Object DeviceID, Size, FreeSpace
$capacity | ConvertTo-Json |
    Set-Content -LiteralPath (Join-Path $directory.FullName 'WIN1-capacity.json') -Encoding utf8
```

Keep the script/output private until reviewed; a path check is not a substitute for approved target/storage selection. Independently compare byte values to the storage UI, inspect the intended output replacement after two runs, and confirm rejected targets/paths do not write an output.

Proposed **read-only APP1** log inspection:

```
journalctl --since today --no-pager
```

Apache request logs have a separate package path. Keep output private. T6 creates/runs no lab automation.

5.4. Expected results

Five events, one explained denial, one safe repeat-run operation and one successful isolated restore with measured limits.

5.5. Verification steps

FWT-07/FWT-08/FWT-09: compare logs to actual actions, automation to another OS view/unchanged configuration, restore output to independent clock/hashes/real HTTPS roles. Keys/backup bytes remain private.

5.6. Troubleshooting

Missing logs: inspect audit/filter/time/rotation. Automation mismatch: inspect target/path/units before retry. Failed restore: preserve original/backup, inspect decryption/storage/version/mode in isolation; do not reconnect to bypass dependencies.

5.7. Rollback procedures

Restore specific changed audit/rotation settings without clearing logs. Remove only intended automation output if approved. Keep copy off/disconnected; resume original only via approved isolation/recovery. Never overwrite sole backup/delete evidence.

6. Phase 5: Validation and Final Submission

6.1. Prerequisites

K/remaining L closeout approval, frozen cases, implementation records, current backup, synthetic targets and private custody. T7 files must exist before final project submission; this does not authorize T7 in T6.

6.2. Required tools

Ten-test catalog, acceptance criteria, clients/policy/log views, evidence index and future T7 files. Git/Python tools support publication only.

6.3. Implementation instructions

1. Freeze expectations and execute every subcheck. Record actual result/time/version/verifier/evidence and second observation. Never change expectation to hide failure.
2. Cover six app cases/revocation, Tailscale permit/deny/revoke, HTTPS, events, automation and restore with bounded lab-owned requests. No scans/protected targets.
3. Mark subchecks PASS, FAIL or NOT TESTED. Timeout without working control/intended denial evidence is insufficient. Preserve failures; corrections/retests need bounded approval.
4. Map ten FWR requirements to evidence/defects. Missing mandatory checks block engineering completion; critical failures block acceptance. Independent human review requires another participant.
5. Complete the five future T7 files below when separately created; own details once and cross-reference. Project final report remains a template until real implementation/reporting is authorized.
6. Review public artifacts/metadata, complete tracked set, staged diff/new history. Keep secrets, addresses, logs, inventories, backups/keys and filled private working records outside Git.
7. Run repository checks, visually inspect changed PDFs, publish branch/PR, verify exact-head CI/review rules, merge with verified noreply identities and fast-forward main. Compare live IDs.
8. Record the owner's explicit workflow UAT reference/date separately. No score/engineering/CI pass supplies acceptance. Stop at scope boundary.

Repository-only T6 checks, not operational tests:

```
python scripts/validate_docs.py --require-finalized
python scripts/validate_office_templates.py
python scripts/build_docs.py --check --rebuild-check
python -m unittest discover -s tests -v
git diff --check
```

6.4. Expected results

Complete evidence, accurate grading, sanitized submission and separate owner decision. T6 delivers documentation/publication only; operational actuals remain empty.

6.5. Verification steps

FWT-10 checks traceability, public hygiene, GitHub/CI and defects. The single rubric totals 100; plans earn no implementation credit and UAT stays PENDING until explicit confirmation.

6.6. Troubleshooting

Correct editable source/build code and regenerate for link/PDF errors; no hand-edited hashes. Satisfy PR reviews/checks without force/bypass. Incomplete UAT stays pending.

6.7. Rollback procedures

Reviewed reverts preserve finalized reports/evidence. End temporary policies/tests only under approved cleanup. Never rewrite history/delete owner work/start a stage to close a report.

7. Five future T7 deliverables

The **approved T7 target, not yet created**, is under `docs/deliverables/`. Existing eight T5 shells remain unchanged and describe the retired baseline; T7 needs separate authorization.

Table 2. Five future T7 deliverables

File	Owning record
`01-Lab-Environment-and-Architecture.xlsx`	Status, four VMs/three zones, flows/capacity
`02-Identity-and-Zero-Trust-Implementation.docx`	AD/groups, endpoints, HTTPS, six cases/app and remote revocation
`03-Security-Monitoring-and-Testing.xlsx`	Five events, investigation, tests, automation/restore
`04-Troubleshooting-and-Lessons.xlsx`	Defects, fixes/retests and lessons
`05-Final-Project-Submission.docx`	Results/index, rubric, limits and separate UAT

8. Historical requirement and test transition

FWR/FWT identify T6. Old R01-R16/T00-T20 keep their meanings in frozen reports/history/T5 shells; no historical PASS transfers to a new operational test. Old quantities below are historical, not current work.

Table 3. Historical requirement and test transition

Earlier requirement	New requirement	Consolidation or retirement
R01	FWR-01	Isolation retained
R02, R03, R04	FWR-02	Capacity/admin; six zones become three
R05, R08	FWR-03	Six standard/two privileged users become three/one; twelve cases become six plus revocation
R06	FWR-04	Windows endpoint/Linux app, no extra endpoint
R07	FWR-05	Two-tier/revocation service becomes one CA/leaf, trust/expiry
R09	FWR-06	Dual pilots become one-resource Tailscale

Earlier requirement	New requirement	Consolidation or retirement
R10	FWR-07	Six sources/twenty-event latency/retention targets become two or three sources/five events/investigation
R12	FWR-08	Two operations become one
R13	FWR-09	Enterprise objectives become measured APP1 restore
R11, R14, R15, R16	FWR-10	Hygiene, focused evidence, governance/submission/UAT retained

Table 4. Historical requirement and test transition

Earlier tests	New test	Treatment
T01, T02, T03	FWT-01; capacity also FWT-02	Isolation/readiness/conditional egress using mocks
T04, T05, T06	FWT-02	Management/permits/default deny/same-zone host controls
T07, T08, T12	FWT-03	AD, six app cases and app revocation
T09	FWT-04	Endpoint baseline
T10, T11	FWT-05	Trust/name/expiry; certificate-revocation test retired
T13, T14	FWT-06	Tailscale only; Twingate retired
T15, T16	FWT-07	Time notes/five events/investigation
T18	FWT-08	One operation plus repeat/rejection
T19	FWT-09	APP1 restore, actual time/limits
T00, T17, T20	FWT-10	Publication/traceability/separate UAT

See decisions, risks and end-to-end coverage. High availability/clustering remain excluded.

9. T6 stop

Finish sources/PDFs, checks, Markdown-only report and reviewed GitHub synchronization. Do not install hardware/software, select disks, create VMs, change networks, deploy AD/enroll Tailscale, run security tests or begin T7. Await owner review.